

Tosion Fashion

A modern full-stack fashion e-commerce platform built with **Next.js**, **TypeScript**, **Express.js**, **MongoDB**, and **Google OAuth**.

Tosion Fashion is designed to provide a clean, editorial-style shopping experience while giving administrators the tools to manage products and content through a dedicated admin side.

🌟 Overview

Tosion Fashion is a modern fashion e-commerce application focused on combining a premium visual experience with practical e-commerce functionality.

The project demonstrates how a production-style fashion platform can be built with a separate frontend and backend architecture.

The platform includes:

- 🛍️ Product browsing and discovery
 - 🔒 Email/password authentication
 - 🔵 Google authentication
 - 👤 Customer accounts
 - 🛠️ Admin product management
 - 🖼️ Product image management
 - 🏷️ Product categories and pricing
 - 📱 Responsive design
 - ⚡ Modern Next.js frontend
 - 🔌 REST API backend
 - 💾 MongoDB database
-

🎯 Why Tosion Fashion?

Many e-commerce projects focus mainly on functionality. Tosion Fashion explores how **functionality and visual presentation can work together**.

The goal is to create an experience that feels closer to a modern fashion brand than a traditional product-management application.

The project is also intended as a practical demonstration of:

- Full-stack TypeScript development
 - Modern Next.js architecture
 - REST API design
 - Authentication and authorization
 - Database integration
 - Admin/customer separation
 - Production-oriented deployment
 - Responsive e-commerce UI
-



Demo



Live demo coming soon / currently under development.

Add your deployed website here when available:

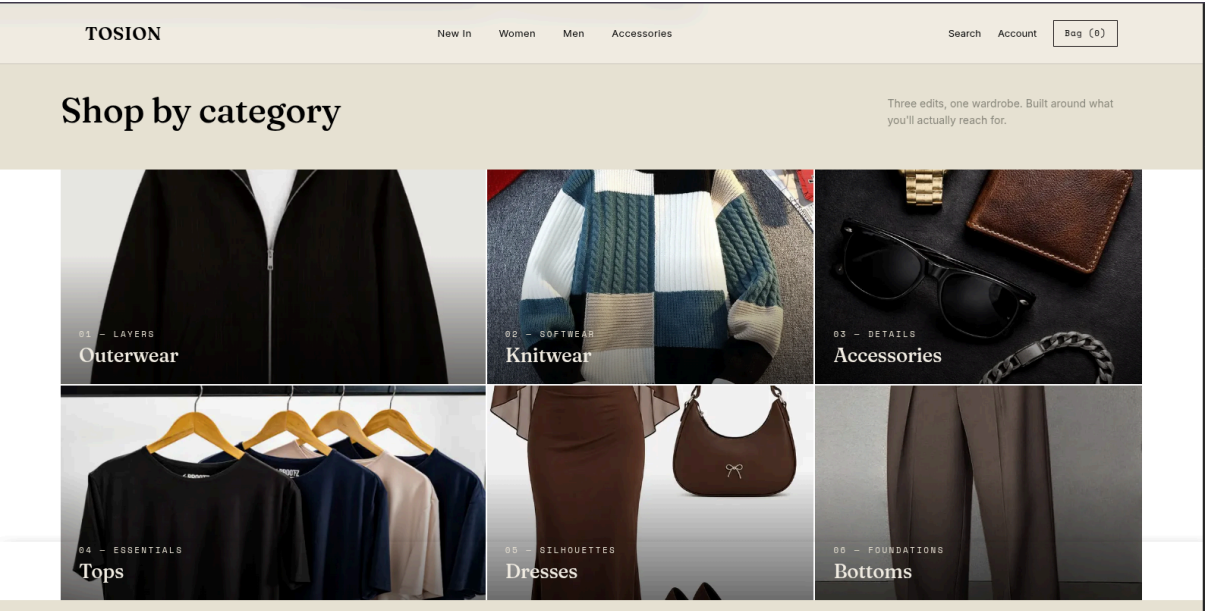
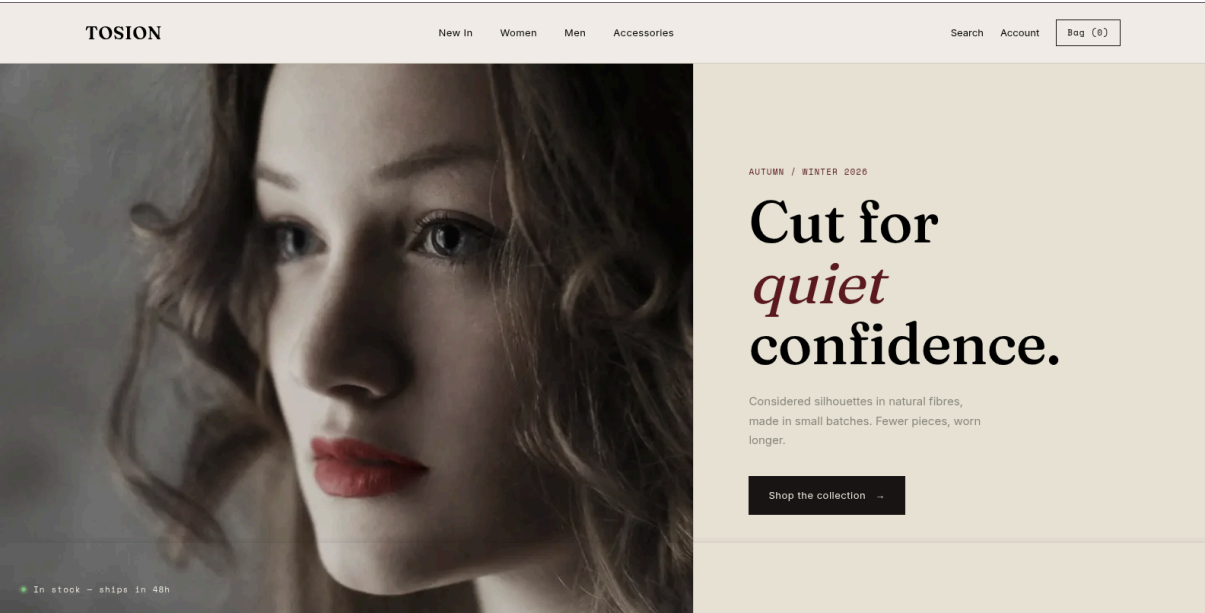
Live Demo: <https://tosion-fashion-livid.vercel.app/>



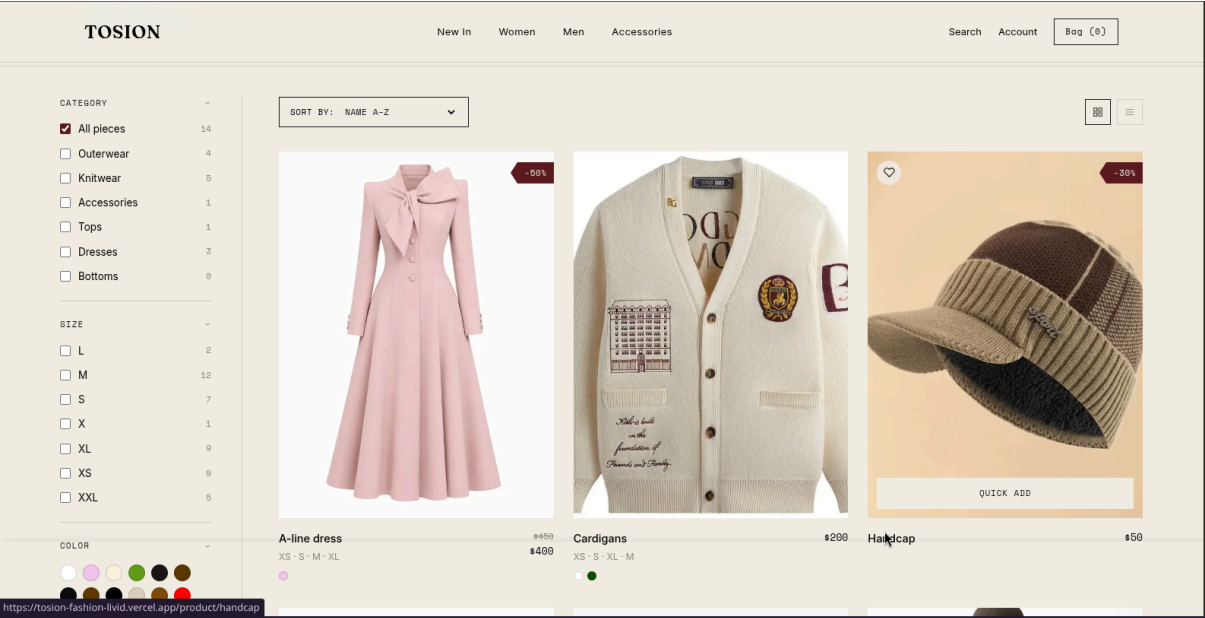
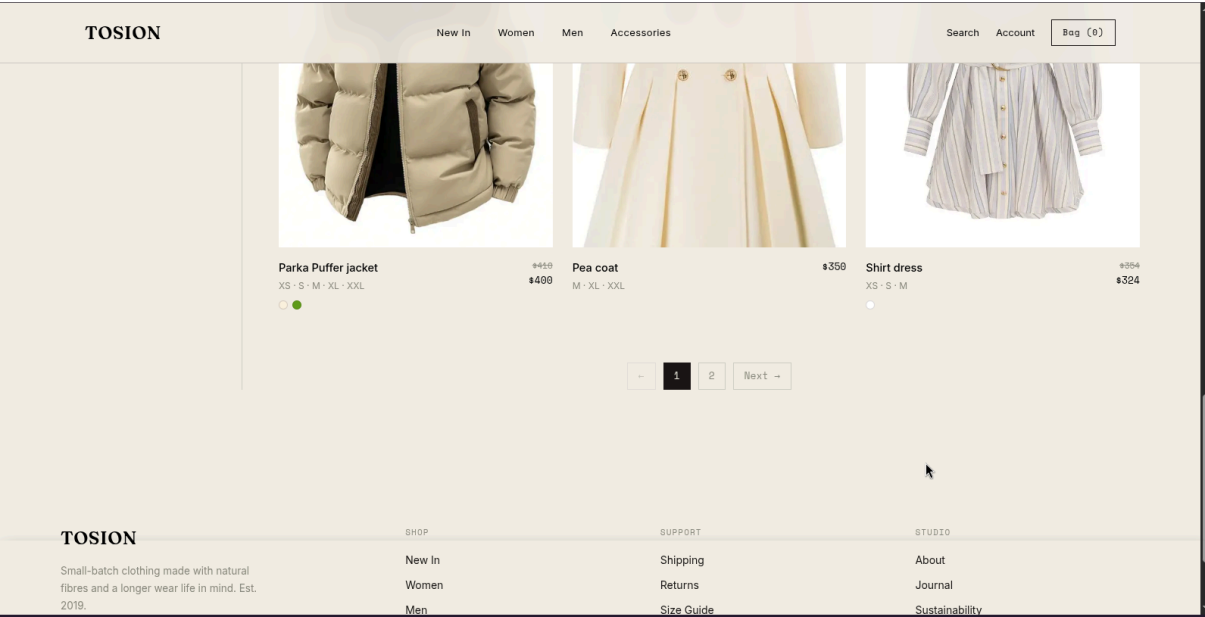
Screenshots

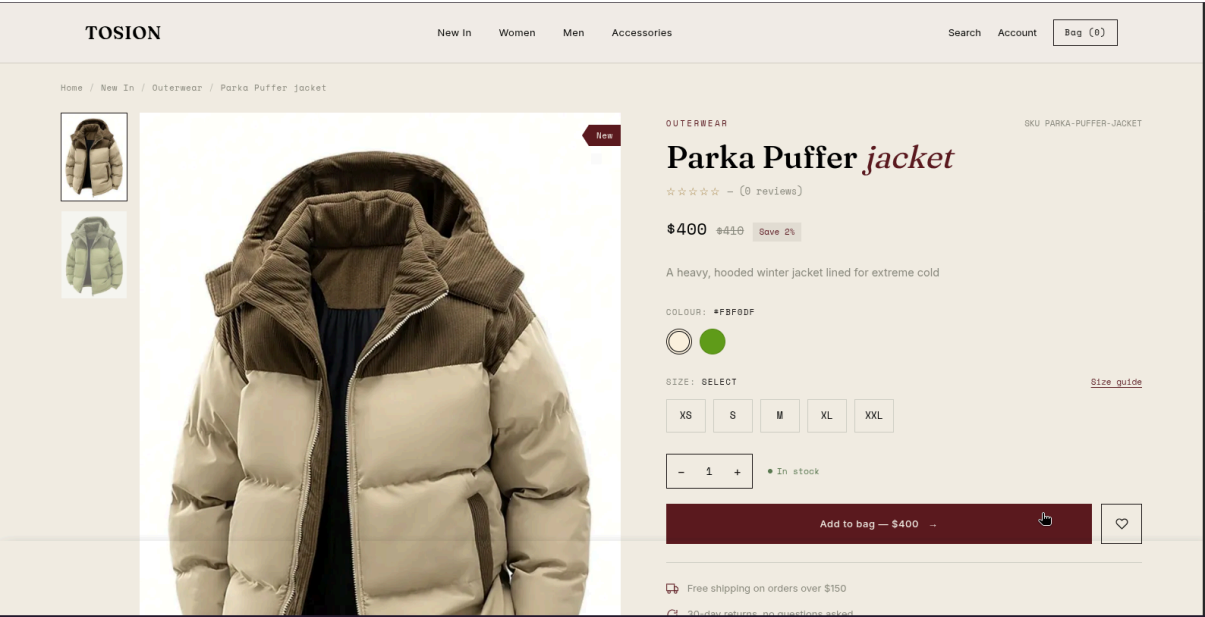
Add screenshots of the main experience here.

Homepage

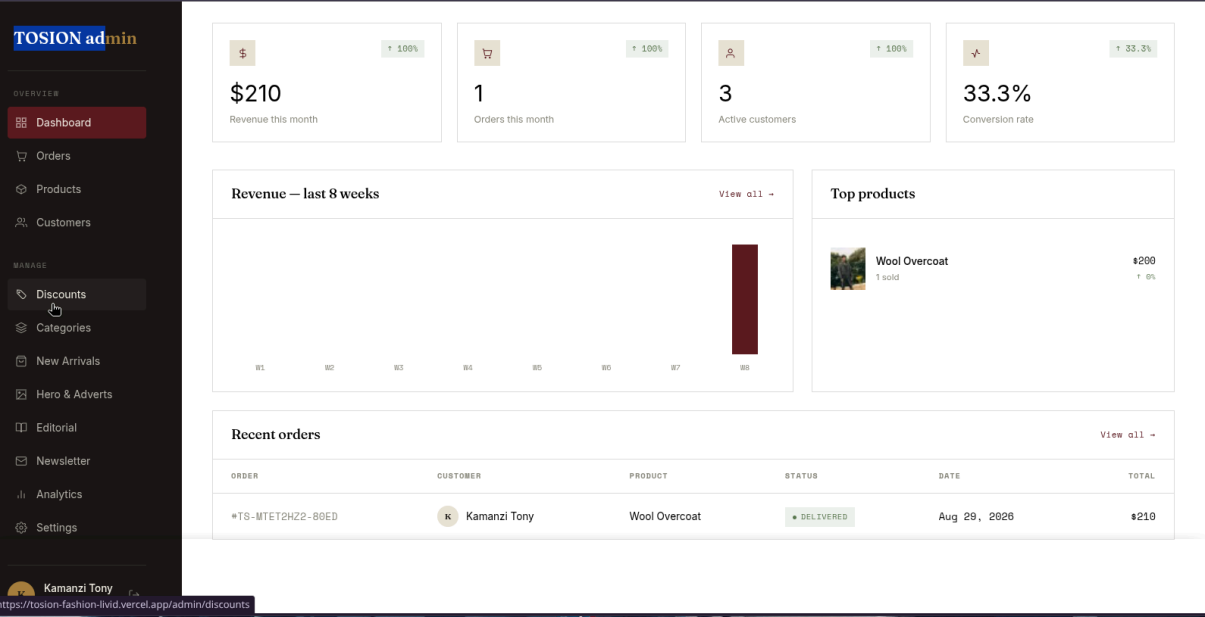


Product Page





Admin Dashboard



Features

Customer

- Browse fashion products
- Explore products by category
- View detailed product information
- Select available sizes

- Create an account
- Sign in with email and password
- Continue with Google
- Manage customer sessions
- Responsive shopping experience

Admin

- Admin authentication
- Product management
- Create products
- Update products
- Delete products
- Manage product pricing
- Manage product categories
- Manage product images

Authentication

Tosion Fashion supports multiple authentication methods:

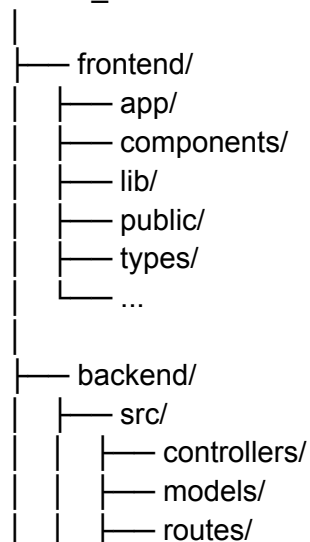
- Email/password authentication
- Google OAuth
- Session-based authentication
- Protected routes
- Customer/admin authorization



Project Architecture

The project is organized into separate frontend and backend applications.

Tosion_fashion/





Frontend

The frontend is responsible for:

- User interface
- Navigation
- Product presentation
- Authentication UI
- Customer experience
- Admin interface
- API communication

Backend

The backend provides:

- REST API
- Authentication
- Authorization
- Product management
- Database communication
- Session management
- Google OAuth integration

Tech Stack

Frontend

- **Next.js**
- **React**
- **TypeScript**
- **Tailwind CSS**
- **Axios**

Backend

- **Node.js**

- **Express.js**
- **TypeScript**
- **Mongoose**
- **Express Session**

Database

- **MongoDB**

Authentication

- Email & password
- **Google OAuth 2.0**
- Session-based authentication

Development & Deployment

- Git
 - GitHub
 - Vercel
 - Render
-



Getting Started

Prerequisites

Before running the project, make sure you have installed:

- Node.js 18+
 - npm
 - MongoDB or a MongoDB Atlas database
 - Git
-

1. Clone the repository

```
git clone https://github.com/TonyKamanzi/Tosion_fashion.git
```

```
cd Tosion_fashion
```

2. Install frontend dependencies


```
cd frontend
```

```
npm install
```

3. Install backend dependencies

Open another terminal:

```
cd backend
```

```
npm install
```

4. Configure environment variables

Create a `.env` file inside the `backend` directory.

Example:

```
PORT=2000
```

```
MONGO_URI=your_mongodb_connection_string
```

```
SESSION_SECRET=your_session_secret
```

```
GOOGLE_CLIENT_ID=your_google_client_id
```

```
GOOGLE_CLIENT_SECRET=your_google_client_secret
```

```
GOOGLE_CALLBACK_URL=http://localhost:2000/auth/google/callback
```

Create the frontend environment file if your frontend requires environment variables:

```
NEXT_PUBLIC_API_URL=http://localhost:2000
```

Never commit your `.env` files or API credentials to GitHub.

5. Start the backend

From the `backend` directory:

npm run dev

The API should now be available at:

http://localhost:2000

6. Start the frontend

From the **frontend** directory:

npm run dev

Open the application at:

http://localhost:3000

Google Authentication

Tosion Fashion supports Google OAuth authentication.

To enable it locally:

1. Create a project in Google Cloud.
2. Configure OAuth credentials.
3. Add your local callback URL.
4. Add the Google credentials to the backend **.env** file.

Example callback URL:

http://localhost:2000/auth/google/callback

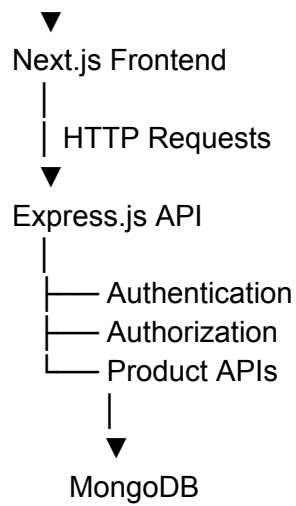
For production, replace the local callback URL with your deployed backend callback URL.

Application Flow

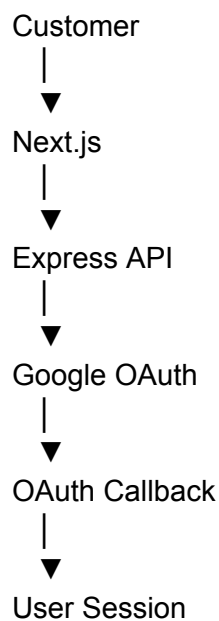
The general authentication flow looks like this:

Customer

|



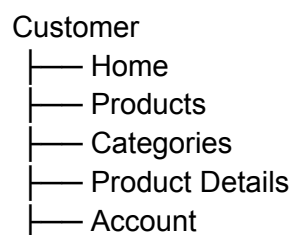
For Google authentication:



Main Application Areas

The application is divided into two primary experiences:

Customer



└─ Authentication

Admin

Admin

└─ Dashboard
└─ Products
└─ Add Product
└─ Edit Product
└─ Product Management

This separation allows the application to provide different functionality and permissions depending on the user's role.



Security Considerations

The project is designed with several common web security practices in mind:

- Environment variables for secrets
- Password authentication
- Session-based authentication
- Protected backend routes
- Role-based access control
- Google OAuth
- CORS configuration
- Server-side validation

Security configuration should be reviewed and hardened further before using the project in a large-scale production environment.



Future Improvements

Tosion Fashion is an evolving project.

Planned improvements include:

- 💳 Payment integration
- 🔍 Advanced product search
- 🎯 Product recommendations
- 📊 Admin analytics
- ✉️ Transactional emails
- 🚚 Order tracking

-  Improved media management
 -  Performance optimization
 -  Automated testing
-

Contributing

Contributions are welcome.

If you have an idea that could improve Tosion Fashion:

1. Fork the repository
2. Create a feature branch

`git checkout -b feature/your-feature`

3. Make your changes
4. Commit your changes

`git commit -m "feat: add your feature"`

5. Push your branch

`git push origin feature/your-feature`

6. Open a Pull Request
-

What This Project Demonstrates

Tosion Fashion is more than a fashion storefront. It is a practical full-stack project demonstrating how different technologies work together to build a modern web application.

It covers:

- Frontend architecture
- Backend architecture
- REST APIs
- Database modeling
- Authentication
- OAuth
- Sessions
- Authorization

- Admin/customer roles
 - Responsive UI
 - API integration
 - Deployment
 - Environment configuration
-



Author

Tony Kamanzi

Full-Stack Developer & Software Development Student

GitHub:

<https://github.com/TonyKamanzi>



License

This project is currently available for learning and development purposes.

★ If you find Tosion Fashion useful or interesting, consider giving the repository a star.